Finding limits — 6.4

Example: $\lim_{x \to 0} \frac{\sin(x)}{x}$ can be evaluated with Taylor series!

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\lim_{x \to 0} \frac{x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7} + \cdots}{x} = \lim_{x \to 0} 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \cdots$$

plugging in $x=0$ gives 1 as the answer.

<u>Example</u>: Compute $\lim\limits_{x\to 0} \dfrac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$.

Recall: $e^x = \sum\limits_{n=0}^{\infty} \dfrac{x^n}{n!} = 1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \dfrac{x^4}{4!} + \dfrac{x^5}{5!} + \cdots$

$$\lim_{x\to 0} \frac{\left(\overbrace{1 + x + \frac{x^2}{2!}} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots\right) \overbrace{- 1 - x - \frac{x^2}{2}}^{\text{cancels}}}{x^3}$$

$$= \lim_{x\to 0} \frac{\frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots}{x^3} = \lim_{x\to 0} \frac{1}{3!} + \frac{x}{4!} + \frac{x^2}{5!} + \cdots = \frac{1}{3!} = \frac{1}{6}.$$